FIELD 24V –> 5V / 100mA – STANDALONE REVIEW CIRCUIT r0.1

All returns are FIELD_GND. This buck is NOT an isolation barrier. No LOGIC_GND connection.
18–30V local–input study; entry chain currently proposes 19–30V. Negative input / retained–output backfeed NOT qualified.

Input effective capacitance >=4,7uF is a qualification gate, not proved by nominal sum. C1–C5 close to VIN/GND.
U1 RT tied to internal VCC selects 1MHz. VCC powers no external load; C6 connects BOOT to SW, never ground.





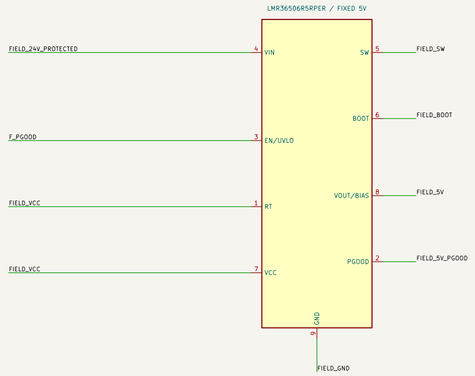




















STARTUP: R1/R2 bias F_PGOOD from protected input.
Entry TPS26630 PGOOD sinks this node until ready.
REMOVE entry R29: no parallel 5V pull–up or DNP path.
J1.3 is an analog power–enable node, not GPIO.

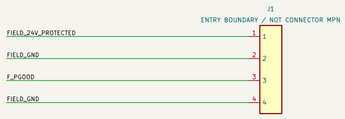








PGOOD below valid VIN is UNKNOWN.
FIELD_5V_PGOOD is diagnostic only.
R4 is a weak bleed; no rapid discharge guarantee.









Release blockers: input/output capacitor derating; fixed–5V PFM accuracy; custom U1/L1 footprints; entry integration; load/inrush; EMC; layout/thermal/bench tests.
Field output loads MUST remain held off through entry ramp and rail startup; driver/inhibit circuit still required. This supply grants no machine permission.
Boundary connectors are schematic interfaces only. No PCB, DRC, Gerbers, assembly placement, isolation rating or physical qualification is claimed.

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